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χ: Symmetry Understanding of 3D Shapes via Chirality Disentanglement

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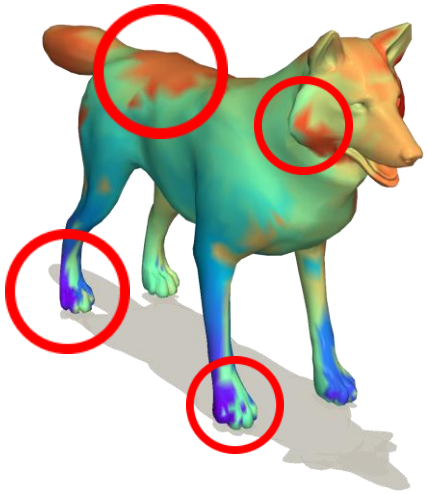
Motivation

Motivation: Generally, 3D shape feature extractors are often limited in their ability to understand chirality, as seen when using Diff3F [1] features to match two shapes (bottom left) and to do 2-set clustering (bottom right):

Source



Diff3F [1]

2-set clustering using Diff3F [1] features

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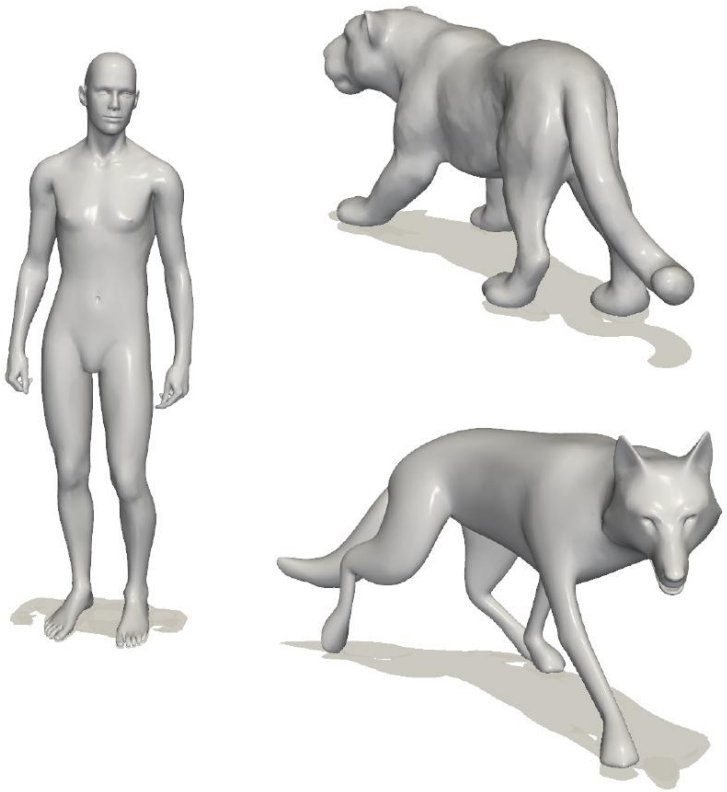
Motivation

Research question: Can we leverage image foundation models to understand symmetries of 3D shapes?



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Contributions

A method that distinguishes left and right of shapes.

- We achieve this by extracting a scalar chirality feature out of shape features computed from image foundation models (DINO and StableDiffusion), in an unsupervised manner.
- We demonstrate that our method outperforms state-of-the-art methods in left-right distinguishment.

Our method effectively solves left-right disambiguity in downstream tasks.

- We show that in shape matching, concatenating our feature with Diff3F [1] features significantly reduces chirality errors.

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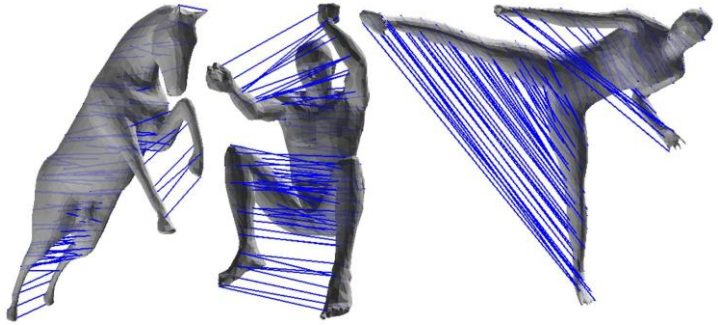
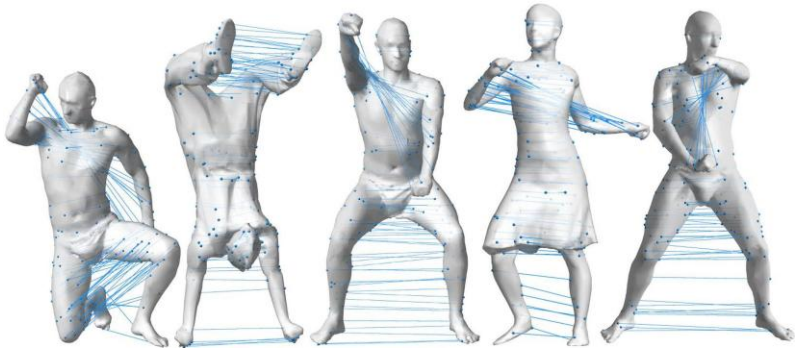
Related Works

- Symmetry detection methods:

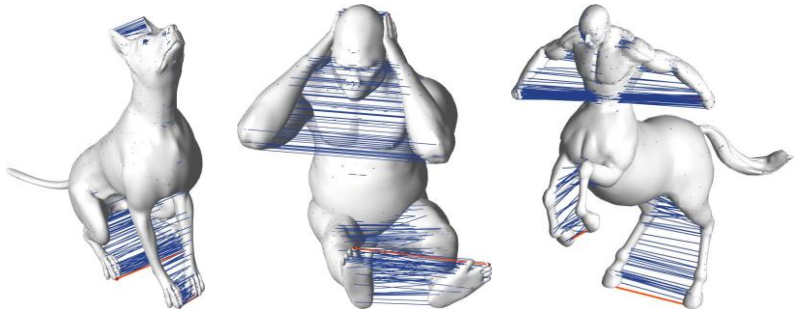
Liu et al. [2]



Kim et al. [4]



Qiao et al. [3]



Nagar et al. [5]

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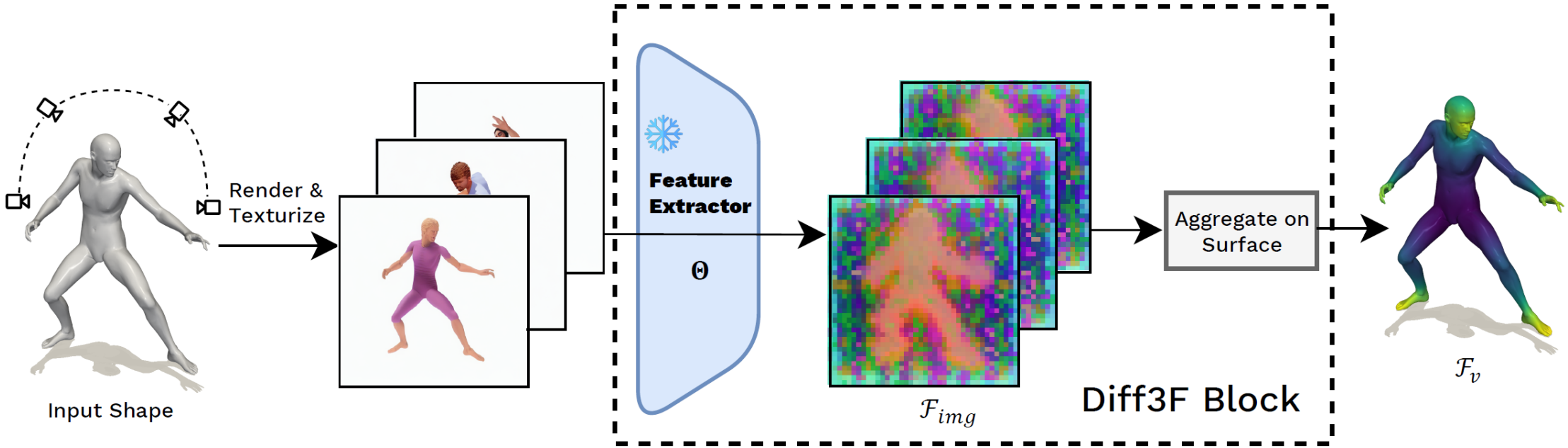
Related Works

- 3D Shape features:



Diff3F [1]

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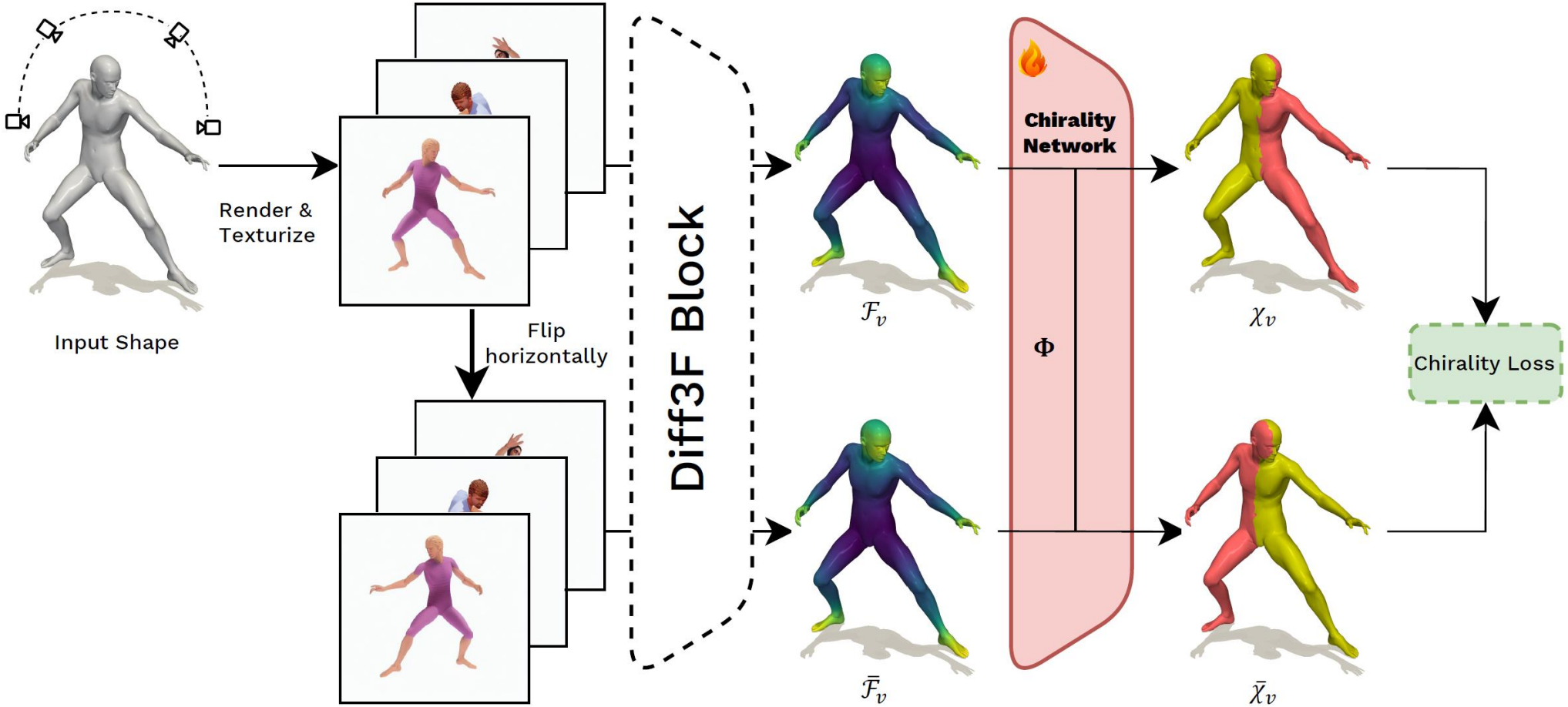
Method



Chirality network: it consists of an encoder g_Φ, a decoder h_Ψ and a projection matrix A.

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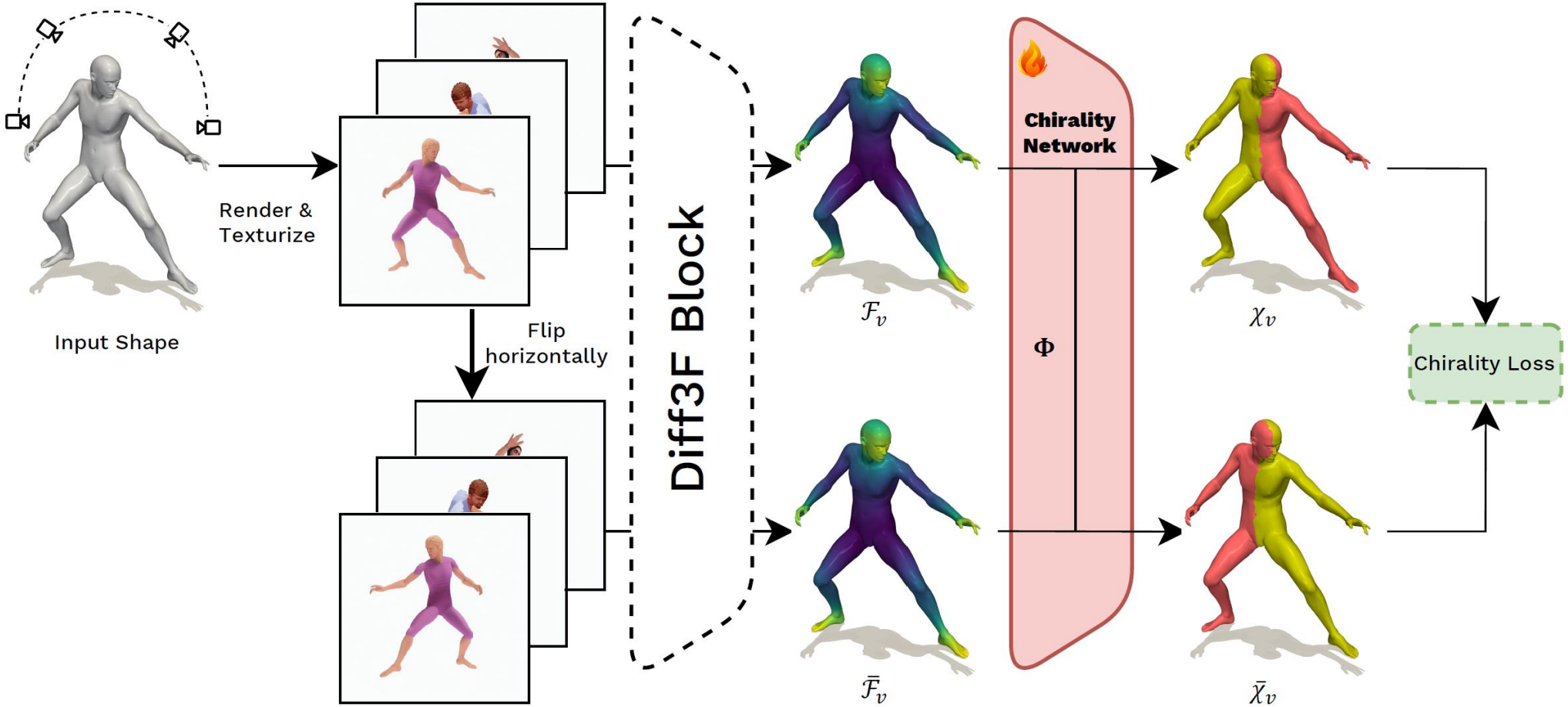
Method



Procedure: For a given shape $\mathcal{X} = (V, E)$, we perform the following steps:

- (1). For each vertex of the shape $v \in V$, we use Diff3F block to generate $\mathcal{F}_v$ and $\bar{\mathcal{F}}_v$ from rendered views and flipped rendered views, respectively, to form feature and semantic-flipped feature pair.

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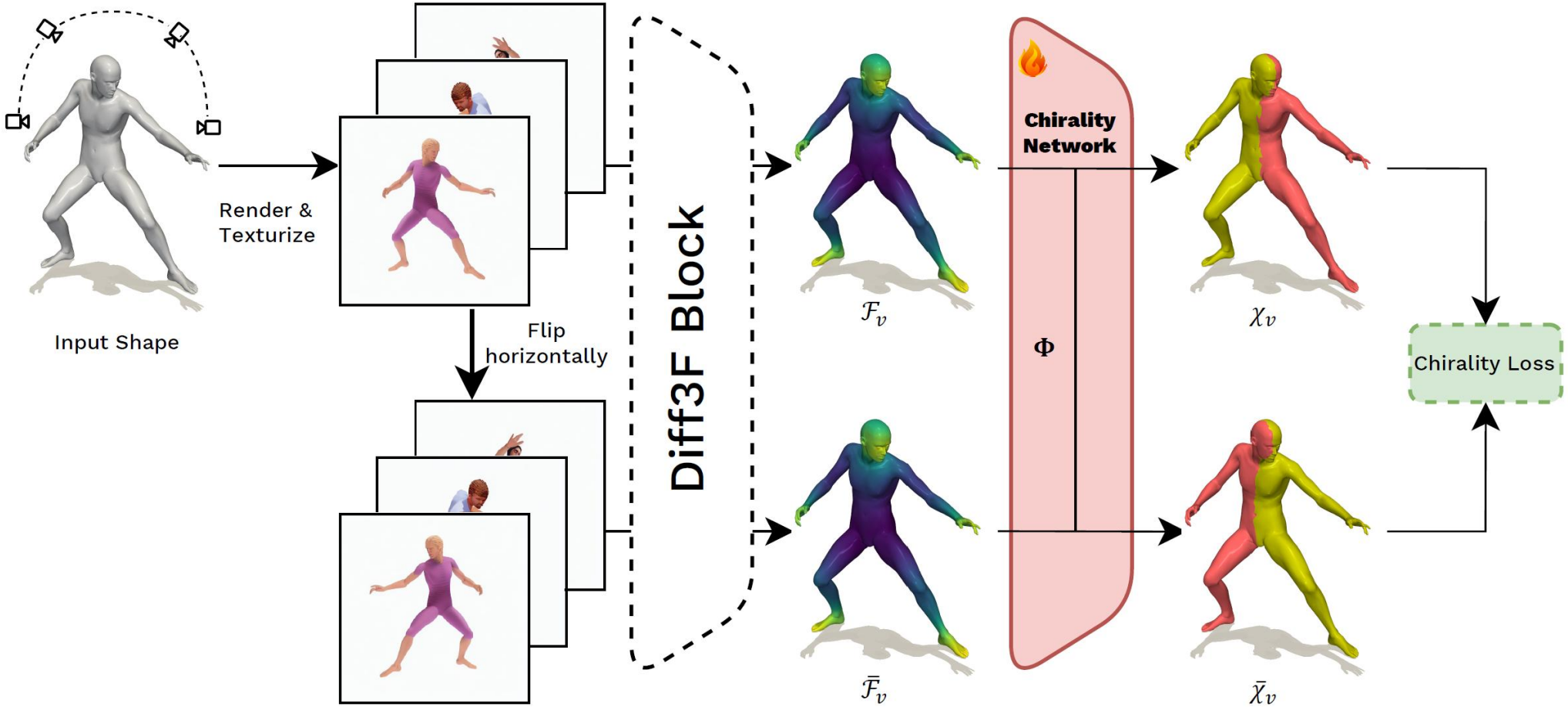
Method



Procedure: For a given shape $\mathcal{X} = (V, E)$, we perform the following steps:

- (2). Compute per-vertex feature $\chi_v = [Ag(\mathcal{F}_v)]_1 / \|Ag(\mathcal{F}_v)\|_2$ and $\bar{\chi}_v = [Ag(\bar{\mathcal{F}}_v)]_1 / \|Ag(\bar{\mathcal{F}}_v)\|_2$, which serve as chirality and semantic-flipped chirality feature, respectively.

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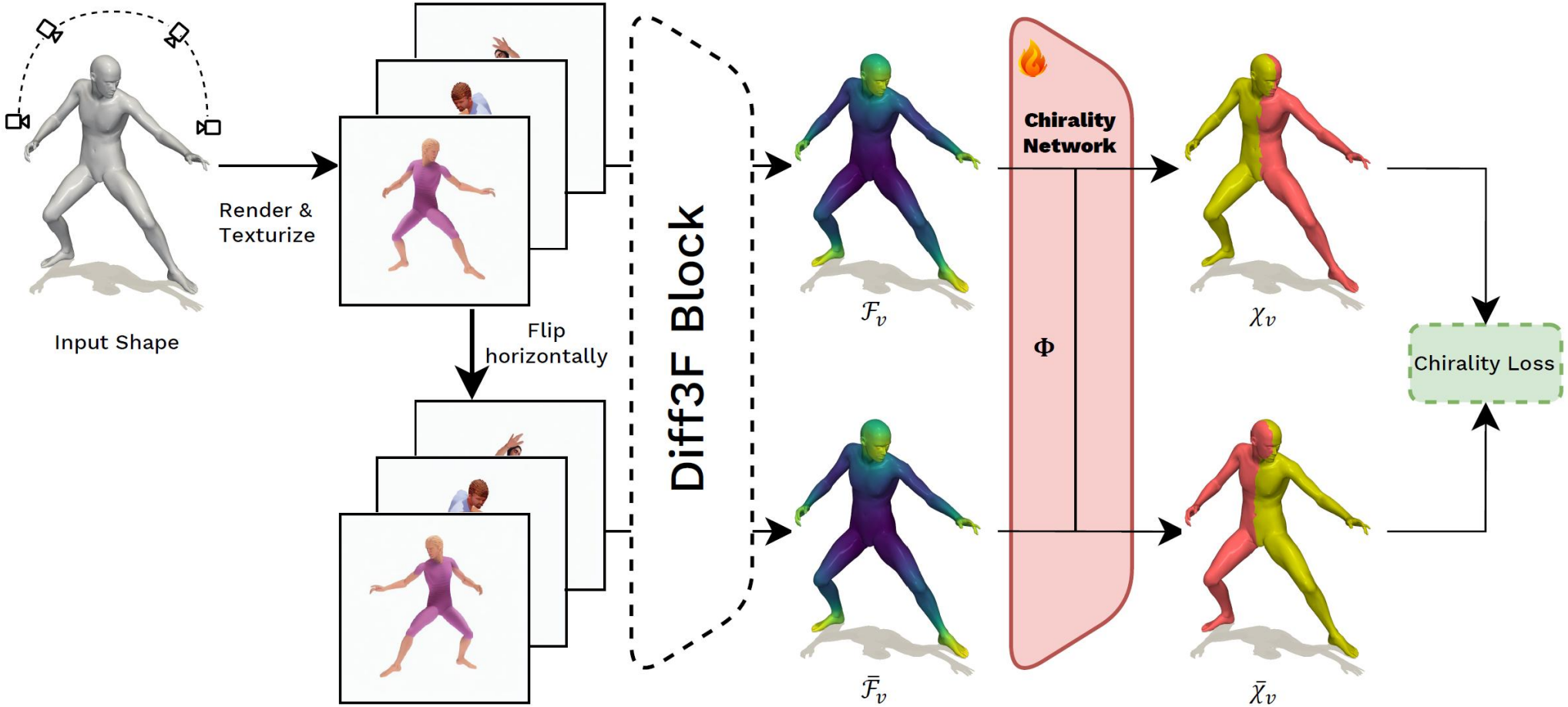
Method



We apply the previous **procedure** to each individual vertex $v \in V$ to form a chirality feature vector χ, and a semantic-flipped chirality feature vector $\bar{\chi}$.

We concatenate $\mathcal{F}_v$ and $\bar{\mathcal{F}}_v$ of all vertices to form feature vectors $\mathcal{F}$ and $\bar{\mathcal{F}}$, respectively.

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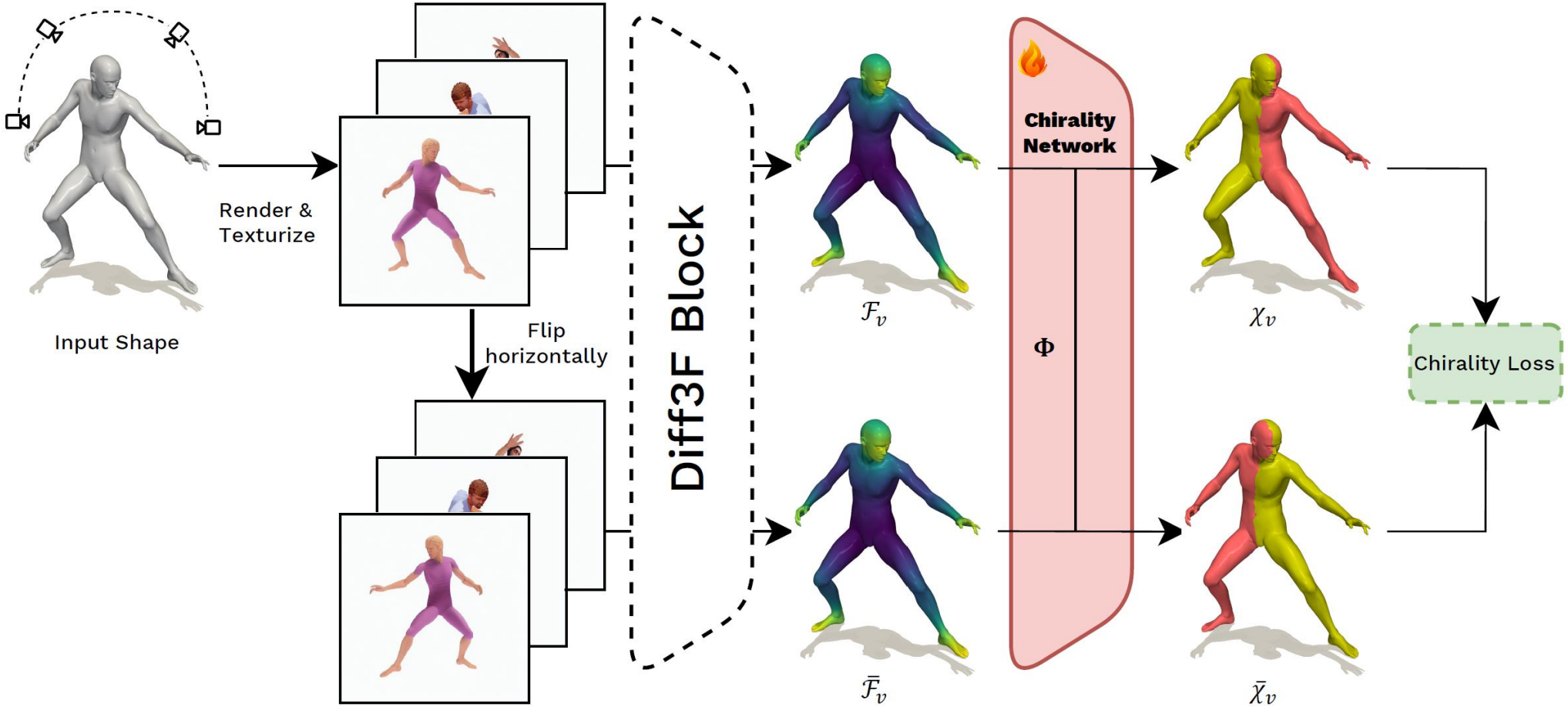
Method



Chirality Losses:

- To maximize the difference between χ and $\bar{\chi}$, we introduce the dissimilarity loss

$$\mathcal{L}_{\text{dis}} = -\frac{1}{|V|}\|\chi - \bar{\chi}\|_2.$$

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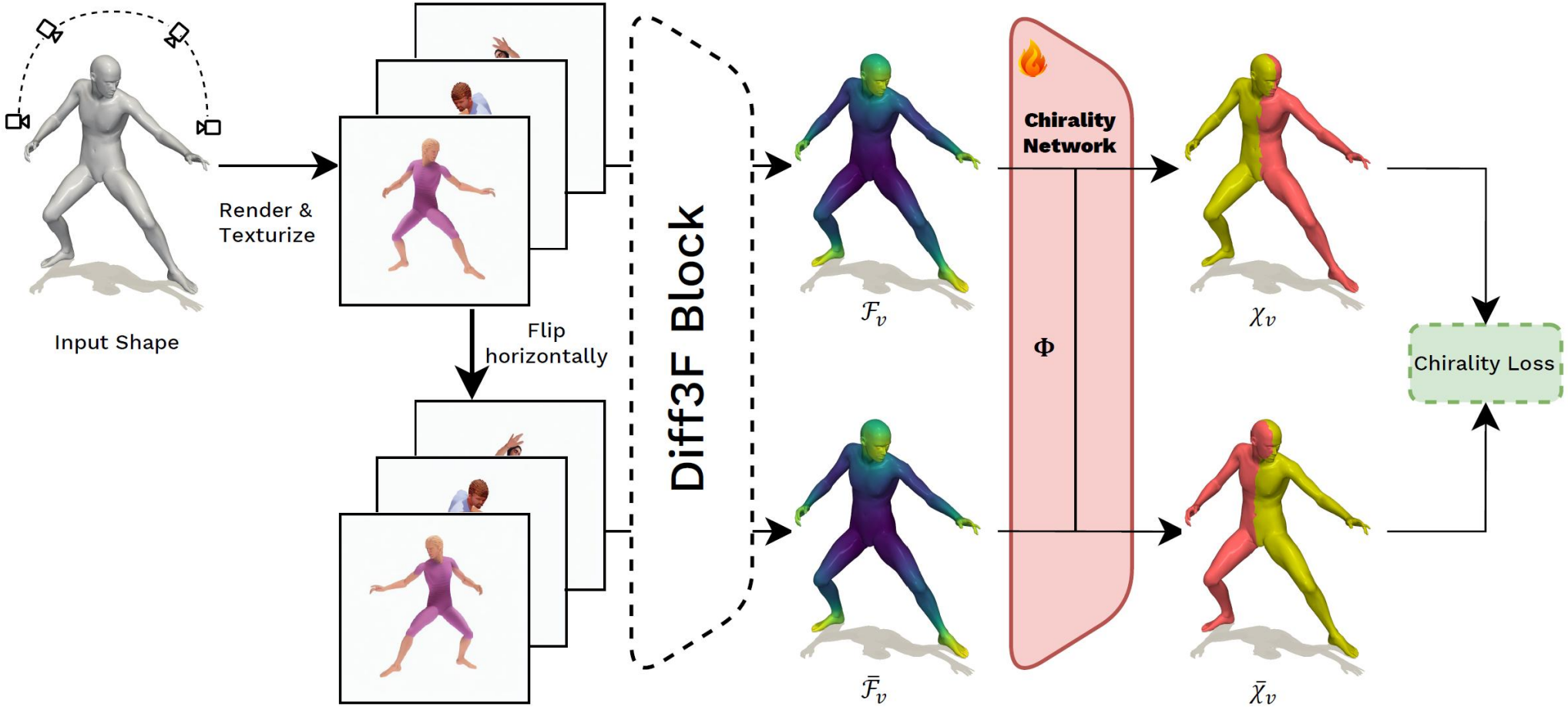
Method



Chirality Losses:

- To achieve spatial smoothness, we introduce the total variation loss

$$\mathcal{L}_{\text{var}} = \frac{1}{|E|} \sum_{(u,v)\in E} \|\chi_u - \chi_v\|_1 + \|\bar{\chi}_u - \bar{\chi}_v\|_1 \,.$$

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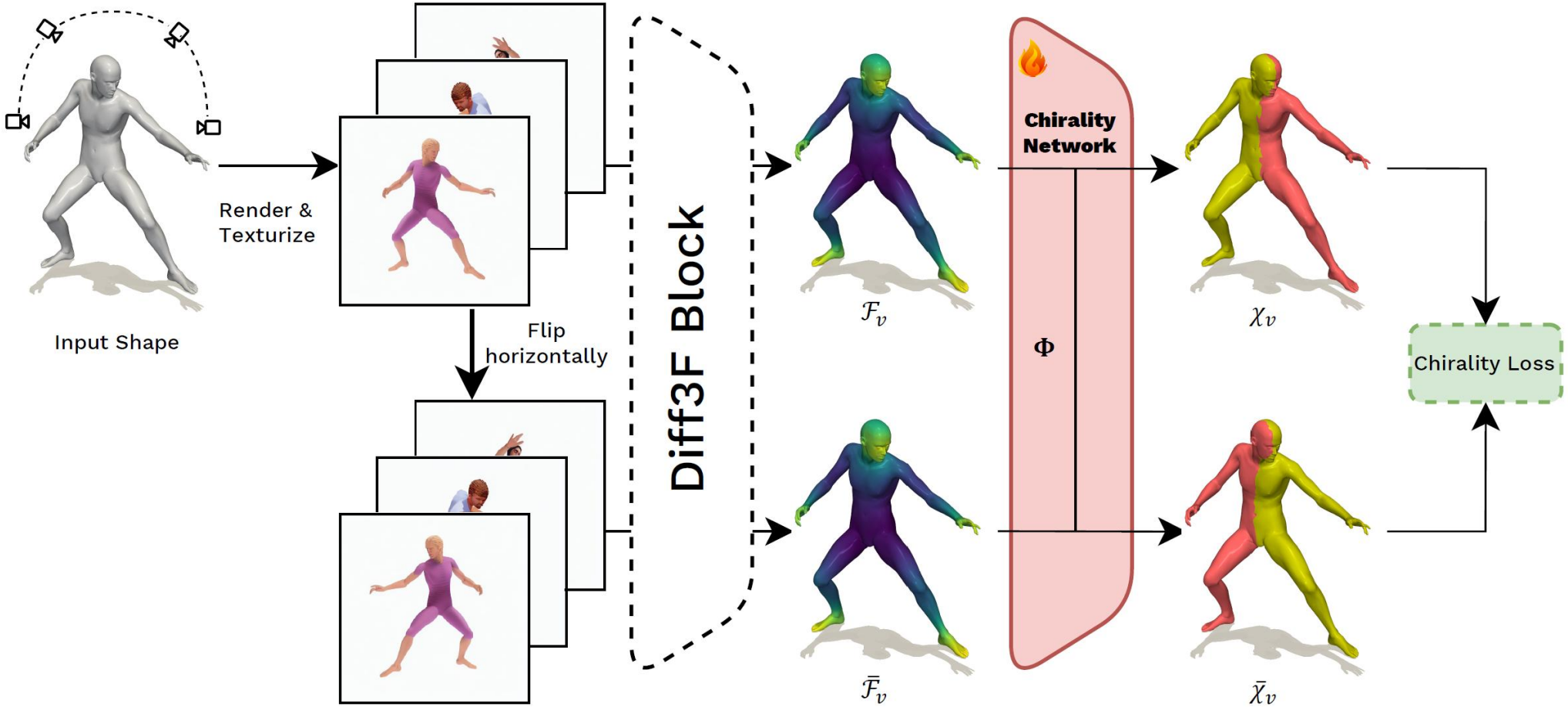
Method



Chirality Losses:

- To stabilize the training process, we include an invertibility loss

$$\mathcal{L}_{\text{inv}} = \frac{1}{\sqrt{|V|}} \| [\mathcal{F}^T \quad \bar{\mathcal{F}}^T]^T - h(g([\mathcal{F}^T \quad \bar{\mathcal{F}}^T]^T)) \|_F .$$

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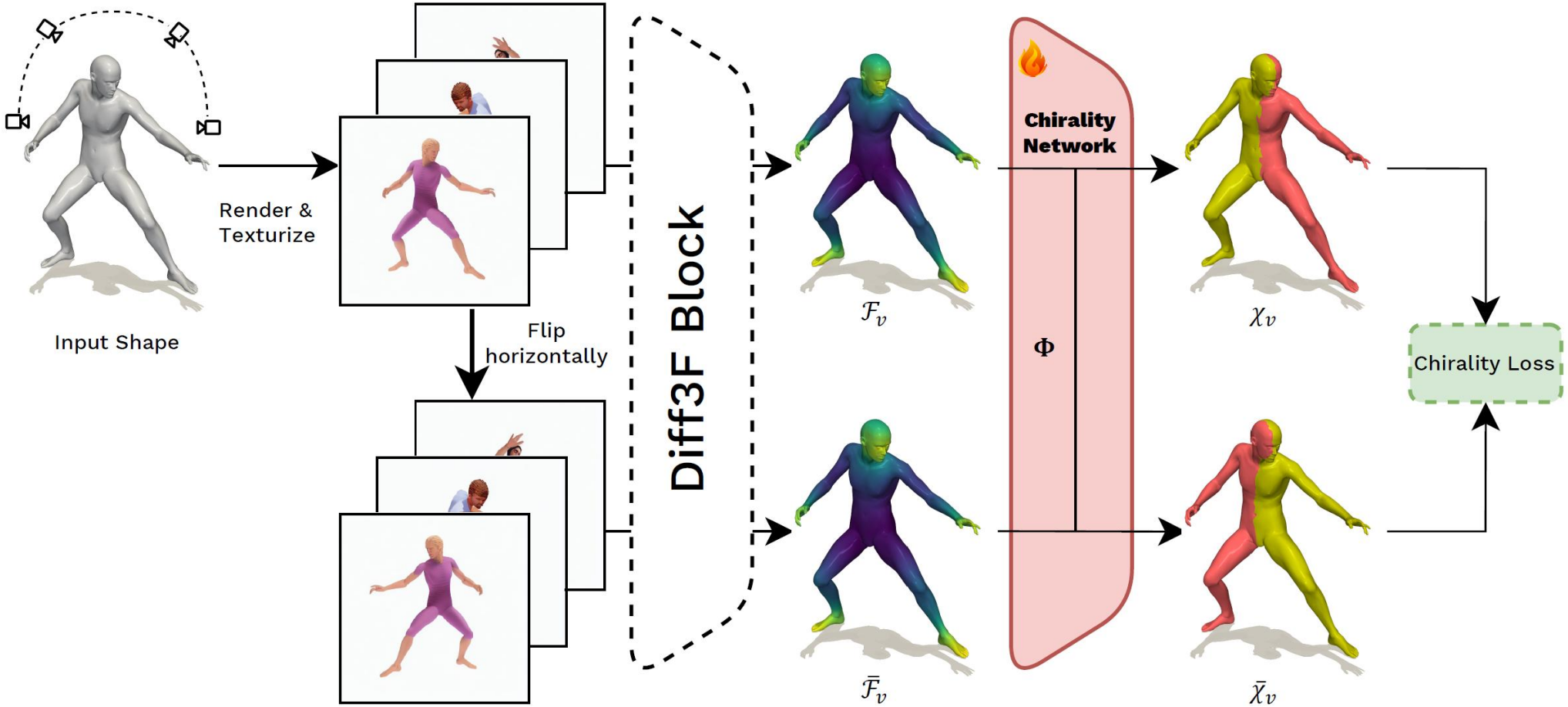
Method



Chirality Losses:

- To make both halves of the shape have the same number of assigned vertices, we introduce a fifty-fifty loss

$$\mathcal{L}_{\text{fif}} = \frac{1}{|V|}\left(\frac{\chi^T \mathbf{1}_{|V|}}{\|\chi\|_\infty} + \frac{\bar{\chi}^T \mathbf{1}_{|V|}}{\|\bar{\chi}\|_\infty}\right).$$

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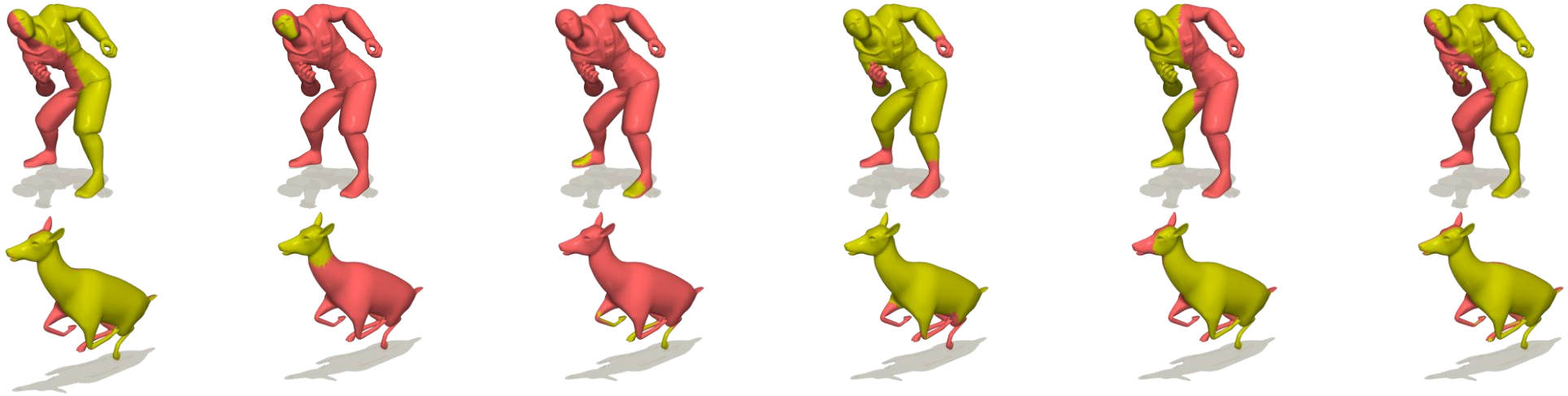
Experiment Results

- Left/Right distinguishment

Our method significantly outperforms its competitors quantitatively and qualitatively.



GT | Diff3F [1] | SD+DINO | Zhang et al. [2] | Liu et al. [3] | Ours

Train	BeCoS	BeCoS-h		BeCoS-a		FAUST		SMAL	
Test	BeCoS	BeCoS-h	BeCoS-a	BeCoS-h	BeCoS-a	FAUST	SCAPE	SMAL	TOSCA
Liu et al. [3]	79.98	79.83	80.46	79.83	80.46	90.45	80.84	75.71	72.88
Ours	91.84	94.09	84.19	90.36	91.10	94.76	95.51	96.59	94.09

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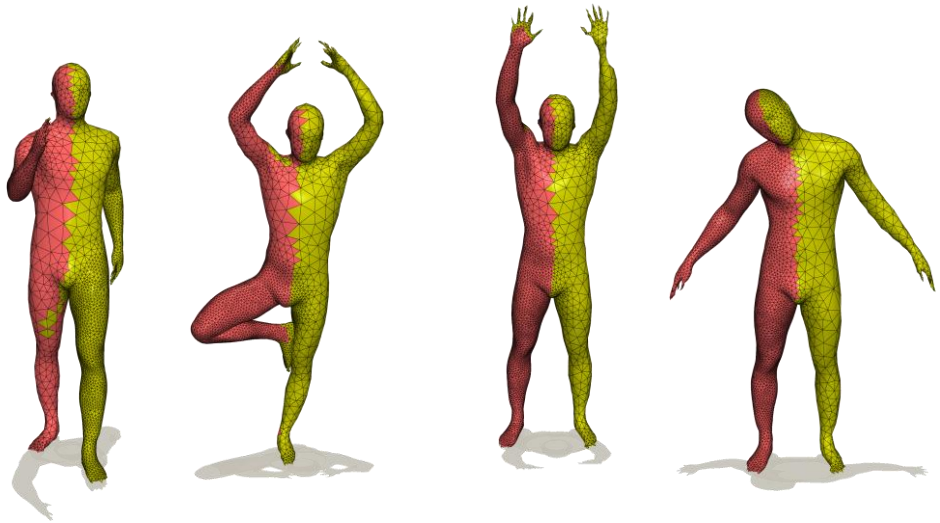
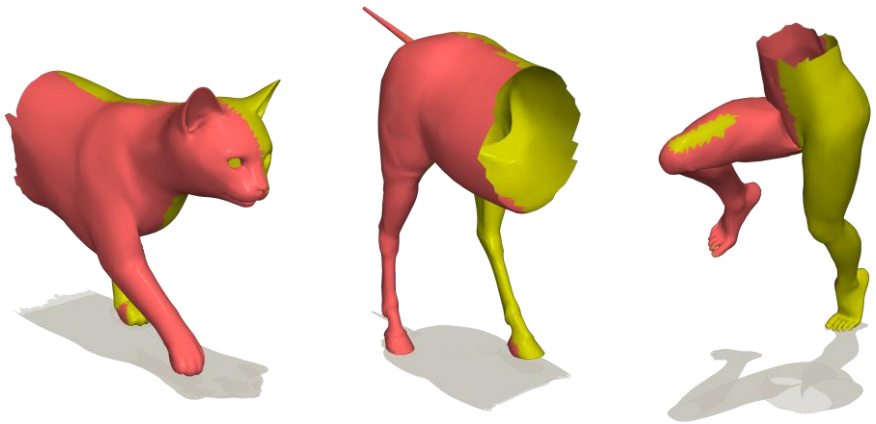
Experiment Results

- Left/Right distinguishment generalisation

We train our model on complete isotropic shapes and test on anisotropic and partial shapes:



Anisotropic shapes



Partial shapes

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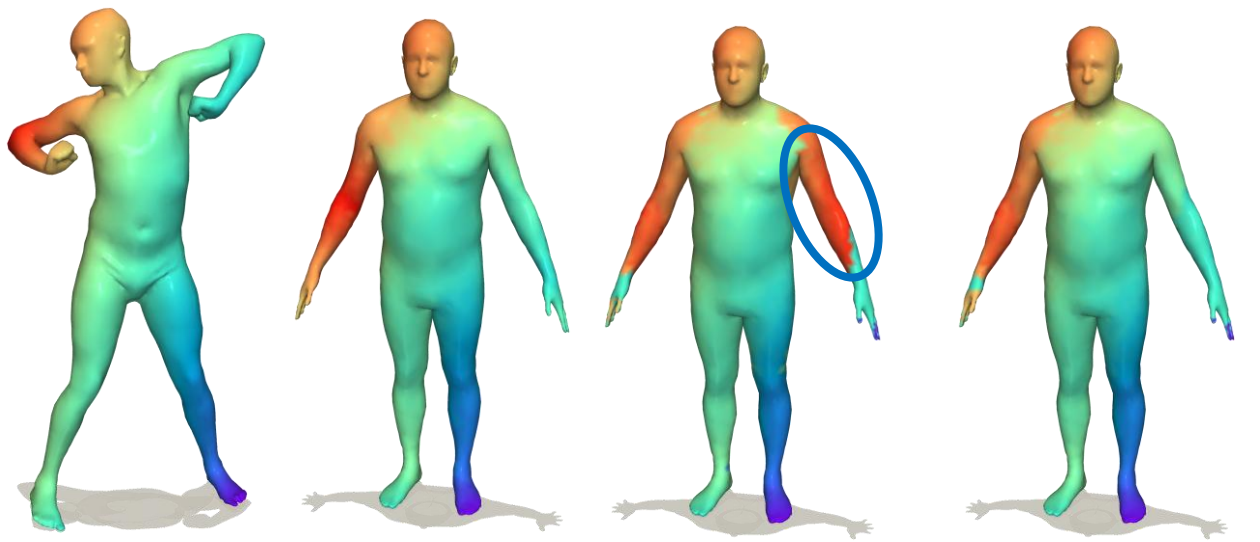
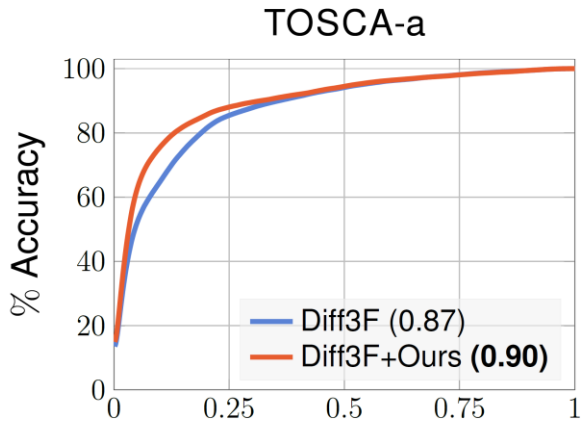
Experiment Results

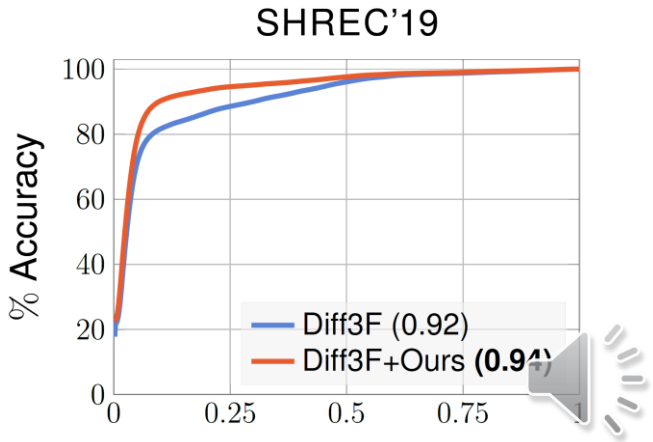
- Shape Matching

When combined with Diff3F features [1], our method effectively resolves left/right ambiguity, as seen in both the qualitative and quantitative results.



Source Target Diff3F [1] Ours+Diff3F





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References

1. Dutt et al. Diffusion 3D Features (Diff3F): Decorating Untextured Shapes with Distilled Semantic Features. CVPR 2024.

2. Liu et al. Finding Surface Correspondences Using Symmetry Axis Curves. Computer Graphics Forum, 2012.

3. Qiao, Yi-Ling, et al. Learning-based intrinsic reflectional symmetry detection. IEEE Transactions on Visualization and Computer Graphics, 2022.

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5. Nagar, Rajendra, and Shanmuganathan Raman. Fast and accurate intrinsic symmetry detection. ECCV, 2018.

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